

# ATHARVA WAGH

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## SUMMARY

Computer Science graduate student with thorough experience in Software Engineering backend development, REST APIs, SQL, cloud infrastructure, and containerized deployments. Strong foundation in C#, C++, JavaScript, TypeScript, object-oriented programming, and data-driven application development. Experienced working with cloud platforms (AWS, exposure to distributed systems) and containerized environments, with a solid understanding of data warehousing, data modeling, and distributed data processing concepts. Proven ability to collaborate with cross-functional teams, contribute to code reviews, documentation, and system optimization, and support data-driven decision-making in production environments.

## TECHNICAL SKILLS

**Languages:** Python, Java, C++, C#, JavaScript, TypeScript, SQL

**Frameworks & Technologies:** React, Node.js, Django, Spring Boot, .NET, Next.js, GraphQL, REST APIs, Vue.js

**Cloud & DevOps:** AWS, Docker, Kubernetes, Linux, Git, CI/CD, Azure DevOps

**Databases & Messaging:** PostgreSQL, MySQL, MongoDB, Redis, Apache Kafka, RabbitMQ

**Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, LLMs, RAG, LangChain

**Core Concepts:** Microservices, Distributed Systems, System Design, Data Pipelines, Parallel Processing, Debugging

## PROFESSIONAL EXPERIENCE

### •Student Software Developer

Los Angeles, CA

(K-12) Viterbi School of Engineering, USC

May 2026 - Present

- Built comprehensive logging and mission style security dashboards with anomaly detection with Azure ML, Splunk & .NET.
- Built scalable backend APIs, orchestration services, and full-stack AI-integrated web modules within enterprise CRM systems.
- Implemented stored procedures in auth pipelines improving operational efficiency and security.
- Contributed to Debugging, System Design, A/B Testing, Backdoor Testing, Privacy Analysis & iterative optimization of production systems in Agile environments.
- Designed scalable logging and monitoring pipelines supporting operational analysis and anomaly detection workflows.

### •Software Developer [Full Time]

Mumbai, India

Institute of Management and Foreign Studies (IMFS)

Jul 2024 - May 2025

- Developed scalable backend services and REST APIs using Python and .NET for cloud-native database-driven applications.
- Designed optimized relational and NoSQL database systems (PostgreSQL, MongoDB) supporting high-performance querying and analytics workflows.
- Worked with AWS cloud infrastructure to deploy, monitor, and maintain production applications and backend services.
- Containerized microservices using Docker & Kubernetes, improving deployment efficiency & application performance by 80%.

### •Web Developer Intern [Co-Op]

Mumbai, India

Mabella SkinCare

Jun 2023 - May 2024

- Engineered a full-stack CRM platform with React, Flask, & PostgreSQL to support high-volume transactional workloads.
- Developed Beautiful Soup Based Web Scraping Service for Market Analysis, supporting inventory & production level decisions
- Conducted debugging and testing of web applications, identifying and resolving issues across frontend and backend layers.
- Designed and optimized database-driven features, improving data retrieval performance and system efficiency.

## PERSONAL PROJECTS

### •NothingDBFromScratch — Relational Database Engine in Modern C++

C++ | CMake | Storage Engine Design | Query Processing | Systems Programming | Linux

- Architected & developed a relational database system from scratch in C++ with modular CMake-based build infrastructure.
- Implemented core database engine concepts including query parsing, execution pipelines, storage/page management, indexing structures, and transaction-processing workflows.
- Explored low-level systems programming concepts including memory-efficient page layouts, B+ tree indexing, buffer management, and join execution strategies.
- Designed compiler-adjacent execution logic involving parsing, optimization planning, execution operators, and runtime-oriented abstractions.
- Built organized scalable multi-module C++ infrastructure using CMake, modular architecture, & reproducible dev workflows.

### •Multi-Agent AI Interview Platform

AWS BedRock | Langchain | UnSloth | Web Scraping | Docker | RAG | Web Sockets | NLP | Apache Kafka | Django

- Architected a multi-agent AI system using LangChain, AWS Bedrock, and Retrieval-Augmented Generation (RAG) pipelines to simulate autonomous technical interview workflows.
- Engineered real-time event-driven communication pipelines using Apache Kafka and WebSockets, enabling low-latency multi-agent orchestration and streaming interactions.
- Integrated tool-calling workflows, web scraping pipelines, vector retrieval systems, and contextual memory mechanisms to improve response quality and agent reasoning.

### •Formula-1 Podium Prediction Model (Pre & Post Quali)

Machine Learning | Predictive Analytics | Python | SQL | AWS SageMaker | Scikit-learn | Statistical Modeling

- Collected, integrated, and analyzed historical and live race telemetry datasets to identify predictive patterns and performance indicators influencing podium outcomes.
- Built scalable analytics pipelines & reusable models supporting efficient querying, feature engineering, & race simulations.
- Applied machine learning and statistical modeling techniques including Logistic Regression, achieving 82.35% true positive rate and 0.98 AUC.

## EDUCATION

### •University of Southern California, Los Angeles, CA

Master of Science in Computer Science

GPA: 3.57/4.0

Aug 2025 - May 2027

### •University of Mumbai, Mumbai, India

Bachelor of Technology in Information Technology (Blockchain Honors)

GPA: 4.0/4.0

Aug 2020 - May 2024